Notes: MATH273 (F25) Basic Theory of Ordinary Differential Equations

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1 Basic Definitions and Examples

Definition 1.1 (Linear and Nonlinear ODEs). We say an ODE is **linear** if

$$F\left[x, y(x), y'(x), \ldots, y^{(n)}(x)\right]$$

depends on $y, y', \ldots, y^{(n)}$ linearly. Note that we allow F to depend on x nonlinearly.

An **nonlinear** ODE is one that is not linear.

We may always represent a linear ODE as

$$a_n(x)y^{(n)}(x) + a_{n-1}(x)y^{(n-1)}(x) + \cdots + a_1(x)y'(x) + a_0(x)y(x) = b(x),$$

where each $a_i(x)$ can be nonlienar in x.

In general, linear ODEs are fully solvable by hand. For nonlinear ODEs, we may only be able to solve some special cases.

2 First Order Linear ODEs

Problem.

$$y'(x) + p(x)y(x) = g(x).$$

2.1 First Order Linear Homogeneous

Problem.

$$y'(t) = a(t)y(t).$$

Integrating, we get

$$\log|y(t)| = \int a(t)\,\mathrm{d}t + C,$$

and so

$$y(t) = \pm e^C \exp\left(\int a(t)\,\mathrm{d}t\right) = C' \exp\left(\int a(t)\,\mathrm{d}t\right),$$

where C' is an arbitrary constant (the case $C' = 0$ is attained when $y = 0$).

2.2 First Order Linear Nonhomogeneous

The Integrating Factor

Problem.

$$y'(x) + p(x)y(x) = g(x).$$

Observe that for each $\mu(t) \neq 0$, the ODE is equivalent to $\mu y' + \mu p y = \mu g$. Using the integrating factor μ:

$$\frac{\mathrm{d}}{\mathrm{d}t}[a(t)y(t)] = a'(t)y(t) + a(t)y'(t) = \mu y' + \mu p y \implies \begin{cases} a = \mu, \\ \mu' = \mu p. \end{cases}$$

A μ is given by the previous case:

$$\mu(t) = \exp\left(\int_{t_0}^{t} p(s)\,\mathrm{d}s\right).$$

We now reduced the ODE to $\frac{\mathrm{d}}{\mathrm{d}t}[\mu(t)y(t)] = \mu(t)g(t)$, which can be solved by integrating and dividing by μ:

$$y(t) = \frac{1}{\mu(t)}\left(\int_{t_0}^{t} \mu(s)g(s)\,\mathrm{d}s + C\right).$$

2.2.1 Separation of Variables

Using the implicit function theorem in principle we can rewrite a first order nonhomogeneous linear ODE as

$$y'(x) = f(x, y)$$

and then

Problem.

$$M(x, y) + N(x, y)y' = 0.$$

For which M, N can we solve this ODE?

(i) The cases

$$M(x) + N(y)y' = 0, \quad \text{or} \quad M_1(x)M_2(y) + N_1(x)N_2(y)y' = 0.$$

are easy: separate and integrate.

(ii) More generally, we can solve

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[\varphi(x, y(x))\right] = g(t) \iff \frac{\mathrm{d}}{\mathrm{d}x}\left[\tilde{\varphi}(x, y(x))\right] = 0$$

using the same trick. When $M + Ny' = 0$ is exact, i.e., if $\partial_2 M = \partial_1 N$, we can turn the ODE into the form above, with

$$\begin{cases} M(x, y) = \partial_1 \varphi(x, y), \\ N(x, y) = \partial_2 \varphi(x, y). \end{cases}$$

In such cases, φ can be found by integrating against each variable.

(iii) Using a integrating factor μ that depends say only on x, we can extend this method to the case where

$$\frac{\mu'}{\mu} = \frac{\partial_2 M - \partial_1 N}{N}$$

is a function of x only.

3 Second Order Linear ODEs

3.1 Homogeneous Second Order Linear ODEs with Constant Coefficients

$$ay'' + by' + cy = 0, \quad a, b, c \in \mathbb{R}.$$

The Ansatz $y(t) = c_0 e^{\lambda t}$ gives

$$a\lambda^2 + b\lambda + c = 0.$$

Three cases:

(i) Two real roots.

$$y(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}.$$

(ii) Two different complex roots $\lambda \pm \mathrm{i}\mu$.

$$y(t) = [c_1 \operatorname{Re} + c_2 \operatorname{Im}] e^{\lambda + \mathrm{i}\mu} = e^{\lambda t} \left[c_1 \cos \mu t + c_2 \sin \mu t\right], \quad c_1, c_2 \in \mathbb{R}.$$

(iii) One real root with a multiplicity two. We use reduction of order with the ansatz $y_2(t) = \mu(t) y_1(t)$. If y_1, y_2 are solutions to $Py'' + Qy' + Ry = 0$ (where P, Q, R are not necessarily constants), then μ sovles

$$\mu'' P y_1 + \mu' \left[2Py_1' + Qy_1\right] = 0 \implies \frac{\mu''}{\mu'} = -\frac{2Py_1' + Qy_1}{Py_1}.$$

3.2 Homogeneous Second Order Linear ODEs with General Coefficients

Problem.

$$Py'' + Qy' + Ry = 0.$$

Ansatz $y(x) = \sum_{n \geq 0}^{\infty} a_n (x - x_0)^n$.

Remark 3.1. This power series is useful for numerical approximation of the solution. We can truncate the series at some N and use the first N terms to approximate the solution.

Theorem 3.2. If $y'' + p(x)y' + q(x)y = 0$ is such that p, q are analytic in $B(x_0, r)$, then there exists a power series solution y to the ODE with convergence radius at least R.

3.3 Nonhomogeneous Second Order Linear ODEs

Problem.

$$Py'' + Qy' + Ry = G.$$

The general solution is

$$y = y_0 + c_1 y_1 + c_2 y_2, \quad \forall c_1, c_2 \in \mathbb{R},$$

where y_0 is a solution to the nonhomogeneous ODE and $c_1 y_1 + c_2 y_2$ is the general solution to the homogeneous ODE. We find y_0 using variation of parameters / constants, with ansatz $y_0 = \mu_1 y_1 + \mu_2 y_2$ and the restriction $\mu_1' y_1 + \mu_2' y_2 = 0$. Then,

$$\begin{aligned} y' &= \mu_1 y_1' + \mu_2 y_2' \\ y'' &= \mu_1' y_1' + \mu_2 y_2' + \mu_1 y_1'' + \mu_2 y_2''. \end{aligned}$$

Plugging into the original ODE,

$$y'' + py' + qy = \mu_1' y_1' + \mu_2' y_2'.$$

Thus we have the 2×2 system

$$\begin{cases} \mu_1' y_1 + \mu_2' y_2 = 0, \\ \mu_1' y_1' + \mu_2' y_2' = g, \end{cases} \iff \begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} \mu_1' \\ \mu_2' \end{pmatrix} = \begin{pmatrix} 0 \\ g \end{pmatrix} \implies \begin{pmatrix} \mu_2 \\ \mu_2 \end{pmatrix}(x) = \int_{x_0}^{x} A^{-1}(t) \begin{pmatrix} 0 \\ g(t) \end{pmatrix} \mathrm{d}t.$$

4 First order ODE system

4.1 The Theory of First Order Homogeneous ODE System

Problem.

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}(t) = \mathbf{A}(t)\mathbf{x}(t).$$

Definition 4.1. We say $\mathbf{x}^1, \ldots, \mathbf{x}^n$ are linear independent if the **Wronski matrix** is not singular, i.e., if the **Wronskian**

$$W(t) := \det\left(x^1, \ldots, x^n\right)$$

is not identically zero.

Theorem 4.2.

$$\frac{\mathrm{d}}{\mathrm{d}t}W(t) = \operatorname{tr}\left(A(t)\right)W(t).$$

Proof. Write

$$M(t) := \begin{pmatrix} x^1(t) & x^2(t) & \ldots & x^n(t) \end{pmatrix} =: \begin{pmatrix} b_1(t) \\ b_2(t) \\ \vdots \\ b_n(t) \end{pmatrix}.$$

Recalling $\frac{\mathrm{d}}{\mathrm{d}t}x^i = Ax^i$ for each i and fixing k, we have $\frac{\mathrm{d}}{\mathrm{d}t}x^i_k = [Ax^i]_k = \sum_j A_{kj}x^i_j$ for each i. Thus by stacking we have

$$\frac{\mathrm{d}}{\mathrm{d}t}b_k = \frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} X^{(1)}_k & \ldots & X^{(n)}_k \end{pmatrix} = \sum_j A_{kj}\begin{pmatrix} X^{(1)}_j & \ldots & X^{(n)}_j \end{pmatrix} = \sum_j A_{kj}b_j.$$

Now,

$$\frac{\mathrm{d}}{\mathrm{d}t}W(t) = \frac{\mathrm{d}}{\mathrm{d}t}\det\begin{pmatrix} b_1 \\ \ldots \\ b_n \end{pmatrix} = \sum_{k=1}^n \det\begin{pmatrix} b_1 \\ \ldots \\ b'_k \\ \ldots \\ b_n \end{pmatrix} = \sum_{k=1}^n \det\begin{pmatrix} b_1 \\ \ldots \\ \sum_j A_{kj}b_j \\ \ldots \\ b_n \end{pmatrix} = \sum_{k=1}^n A_{kk}\det\begin{pmatrix} b_1 \\ \ldots \\ b_k \\ \ldots \\ b_n \end{pmatrix} = \sum_k A_{kk}W(t).$$

□

Corollary 4.3. Either $W \equiv 0$ or $W(t) \neq 0$ for each t. In particular, it suffices to check $W(t_0)$ at any t_0 to determine linear independence.

Proof. Note that

$$W(t) = W(t_0) \exp\left(\int_{t_0}^{t} \operatorname{tr}(A(s))\, \mathrm{d}s\right),$$

where the last term is never zero. □

Theorem 4.4. Suppose that $x^1, \ldots, x^n$ are n linearly independent *real* solutions to the homogeneous ODE system $\frac{\mathrm{d}}{\mathrm{d}t}x = A(t)x$. Then any solution x to the ODE can be written uniquely as

$$x(t) = \sum_{j=1}^{n} c_j x^j(t), \quad c_j \in \mathbb{R}.$$

In particular, the space of solutions to the homogeneous ODE system (or any nth order ODE) is an at most n-dimensional vector space.

Proof. For each $c \in \mathbb{R}^n$ write $y_c(t) = x(t) - \sum_i c_i x^i(t)$. By previous results, y_c solves the homogeneous ODE.

Now fix t_0, we seek c_i such that

$$x(t_0) = \begin{pmatrix} x^1(t_0) & x^2(t_0) & \ldots & x^n(t_0) \end{pmatrix} c =: M(t_0)c.$$

Since $x^1, \ldots, x^n$ are linearly independent, the Wronski matrix M is invertible, and we can obtain c as $M^{-1}(t_0)x(t_0)$.

In particular we have $y_c(t_0) = 0$ and y_c solves the homogeneous ODE. Since 0 is also a solution to the homogeneous ODE, by uniqueness we have $y_c(t) \equiv 0$, which gives $x(t) \equiv \sum c_i x^i$. □

Theorem 4.5. Suppose that x^i solves $\frac{\mathrm{d}}{\mathrm{d}t}x^i = A(t)x^i$ with the IC $x^i(t_0) = e_i$ for $1 \le i \le n$. Then x^i are linearly independent solutions.

In particular, the solution space is at least n-dimensional.

Proof. The existence and uniqueness theorem guarantees that x^i exist and are unique. To check that they are independent, note that $W(t_0) = \det I = 1 \neq 0$. But note of course that we can pick any n linearly independent initial conditions in $\mathbb{R}^n$. □

4.2 First Order Linear Homogeneous ODE System with Constant Coefficients

4.2.1 Eigenpairs

Problem.

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = Ax(t), \quad A \in \mathbb{R}^{n\times n}.$$

The ansatz $x(t) = e^{\lambda t} v$, $v \in \mathbb{R}^{n\times 1}$, $\lambda \in \mathbb{R}$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(e^{\lambda t}\mathbf{v}\right) = \lambda e^{\lambda t}\mathbf{v}, \quad A(e^{\lambda t}\mathbf{v}) = e^{\lambda t} A\mathbf{v}.$$

Thus $\mathbf{x}(t)$ solves the ODE system if and only if $\lambda v = Av$, or if and only if (λ, v) is an eigenpair of A. After projecting in the direction of the eigenvectors, the ODE system decouples.

A few cases:

(i) A has n distinct real eigenvalues (so the eigenvectors are real and linearly independent). Since $W(0) = \det\left(v_1, \ldots, v_n\right) \neq 0$, we have n linearly independent solutions:

$$y = c_1 e^{\lambda_1 t} v_1 + \ldots c_n e^{\lambda_n t} v_n.$$

(ii) More generally, A has n linearly independent real eigenvectors (or equivalently, A is diagonalizable). [1]

(iii) A has complex eigenvalues, but all eigenvalues are distinct. Note that if (λ, v) is an eigenpair, then so is $(\overline{\lambda}, \overline{v})$, since $A\mathbf{v} = \lambda\mathbf{v} \iff A\overline{\mathbf{v}} = \overline{\lambda}\overline{\mathbf{v}}$ (Note that this relies on A being real). We seek two real solutions from $e^{\lambda t} v$ and $e^{\overline{\lambda} t}\overline{v}$. Suppose then that $\lambda = \alpha + \mathrm{i}\beta$ and $v = a + \mathrm{i}b$ for $\alpha, \beta \in \mathbb{R}$, $a, b \in \mathbb{R}^{n\times 1}$. Then

$$\begin{aligned} e^{\lambda t} v &= e^{(\alpha+\mathrm{i}\beta)t}(a + \mathrm{i}b) \\ &= e^{\alpha t}(\cos(\beta t) + \mathrm{i}\sin(\beta t))(a + \mathrm{i}b) \\ &= e^{\alpha t}\left[(\cos(\beta t)a - \sin(\beta t)b) + \mathrm{i}(\sin(\beta t)a + \cos(\beta t)b)\right] \end{aligned}$$

and similarly,

$$e^{\overline{\lambda} t}\overline{v} = e^{\alpha t}\left[(\cos(\beta t)a - \sin(\beta t)b) - \mathrm{i}(\sin(\beta t)a + \cos(\beta t)b)\right].$$

From this we see that

$$\begin{aligned} \operatorname{Re} e^{\lambda t} v &= \frac{e^{\lambda t} v + e^{\overline{\lambda} t}\overline{v}}{2} = e^{\alpha t}(\cos(\beta t)a - \sin(\beta t)b) \\ \operatorname{Im} e^{\lambda t} v &= \frac{e^{\lambda t} v - e^{\overline{\lambda} t}\overline{v}}{2\mathrm{i}} = e^{\alpha t}(\sin(\beta t)a + \cos(\beta t)b) \end{aligned}$$

are two real solutions to the ODE system.

Thus each complex eigenpair gives two real solutions to the ODE system.

[1] This is a strict superset of the previous case, think

$$A = \begin{pmatrix} 1 & & \\ & 1 & \\ & & 2 \end{pmatrix}.$$

4.2.2 Matrix Exponential

Problem.

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = Ax(t), \quad A \in \mathbb{R}^{n\times n}.$$

$$\exp[A] \coloneqq \sum_{k=0}^{\infty} \frac{A^k}{k!} \in \mathbb{R}^{n\times n}.$$

Lemma 4.6.

(i) If $B = T^{-1}AT$, then $\exp[B] = T^{-1}\exp[A]T$.

(ii) If $AB = BA$, then $\exp[A+B] = \exp[A]\cdot\exp[B]$.

(iii) $(\exp[A])^{-1} = \exp[-A]$.

Proof.

(i) Note only that $B^k = T^{-1}A^kT$.

(ii) Note that

$$\begin{aligned}
\exp[A+B] &= \sum_{k=0}^{\infty} \frac{(A+B)^k}{k!} = \sum_{k\geq 0} \frac{1}{k!}\left(\sum_j \binom{k}{j} A^j B^{k-j}\right) \\
&= \sum_{k=0}^{\infty}\sum_{j=0}^{k} \frac{A^jB^{k-j}}{j!(k-j)!} = \sum_{j=0}^{\infty}\sum_{k=j}^{\infty} \frac{A^j}{j!}\frac{B^{k-j}}{(k-j)!} = \sum_{j=0}^{\infty}\sum_{k=0}^{\infty} \frac{A^j}{j!}\frac{B^k}{k!} \\
&= \exp[A]\exp[B].
\end{aligned}$$

(iii) Applying property (ii) with $B = -A$ gives $\exp[A]\exp[-A] = \exp[0] = I$. Similarly, $\exp[-A]\exp[A] = I$.

□

Proposition 4.7. Suppose $Av = \lambda v$. Then $\exp[A]v = e^{\lambda}v$.

Proof. Use the fact that $A^kv = \lambda^k v$. □

Proposition 4.8.

$$\frac{\mathrm{d}}{\mathrm{d}t}\exp[tA] = A\exp[tA] = \exp[tA]A.$$

Proof. Note that $t \mapsto \exp[tA]$ behaves like a power series with radius of convergence ∞. Term-by-term differentiation gives

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left(\sum_{k=0}^{\infty}\frac{(tA)^k}{k!}\right) &= \sum_{k=0}^{\infty}\frac{\mathrm{d}}{\mathrm{d}t}\frac{(tA)^k}{k!} = \sum_{k=1}^{\infty}\frac{A^k}{k!}kt^{k-1} = \sum_{k\geq 1}\frac{A^k}{(k-1)!}t^{k-1} \\ &= A\sum_{k=1}^{\infty}\frac{(tA)^{k-1}}{(k-1)!} = A\exp[tA].\end{aligned}$$

But in the last line we may as well place A at the end to obtain $\frac{\mathrm{d}}{\mathrm{d}t} = \exp[tA]A$. □

Theorem 4.9. Suppose that $A \in \mathbb{R}^{n\times n}$. Then the solution to $\frac{\mathrm{d}}{\mathrm{d}t}x = Ax, \quad x(t_0) = x_0 \in \mathbb{R}^{n\times 1}$ is given by

$$x(t) = e^{A(t-t_0)}x_0.$$

Proof. Note that

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = \frac{\mathrm{d}}{\mathrm{d}t}\left(e^{A(t-t_0)}x_0\right) = Ae^{A(t-t_0)}x_0 = Ax(t).$$

At $t = t_0$ we have

$$x(t_0) = e^{A\cdot 0}x_0 = Ix_0 = x_0.$$

By the existence and uniqueness theorem, this is the unique solution. □

Remark 4.10.

$$\exp[Nt] = I + Nt + \frac{(Nt)^2}{2!} + \cdots + \frac{(Nt)^{l-1}}{(l-1)!} \qquad = \begin{pmatrix} 1 & t & \frac{t^2}{2!} & \dots & \frac{t^{l-1}}{(l-1)!} \\ 0 & 1 & t & \dots & \frac{t^{l-2}}{(l-2)!} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$

Example 4.11.

$$\frac{\mathrm{d}}{\mathrm{d}t}x = \begin{pmatrix} \lambda & 1 & \\ & \lambda & 1 \\ & & \lambda \end{pmatrix}x$$

has solution

$$e^{At} = e^{\lambda t}e^{Nt} = e^{\lambda t}\begin{pmatrix} 1 & t & \frac{t^2}{2} \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}.$$

4.3 Nonhomogeneous First Order Linear ODE Systems

Problem.

$$\frac{\mathrm{d}}{\mathrm{d}t}x = A(t)x + G(t), \quad x(t_0) = x_0.$$

First recall the case $n = 1$

$$\frac{\mathrm{d}}{\mathrm{d}t}x = ax + g(t), \quad a \in \mathbb{R}.$$

Using the integrating factor e^{-at}, we have

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(e^{-at}x\right) = e^{-at}g(t)$$

and so

$$e^{-at}x(t) = e^{-at_0}x(t_0) + \int_{t_0}^{t} e^{-as}g(s)\,\mathrm{d}s,$$

giving

$$\begin{aligned} x(t) &= e^{a(t-t_0)}x_0 + \int_{t_0}^{t} e^{a(t-s)}g(s)\,\mathrm{d}s \\ &= e^{a(t-t_0)}\left[x_0 + \int_{t_0}^{t} e^{-a(s-t_0)}g(s)\,\mathrm{d}s\right]. \end{aligned}$$

Note that the first term $e^{a(t-t_0)}x_0$ solves the homogeneous ODE with the given IC, and for each s, the term $e^{a(t-s)}g(s)$ solves the homogeneous ODE with IC $g(s)$ at $t = s$. We may think of them as solutions to the following two ODEs:

$$\begin{cases} \frac{\mathrm{d}}{\mathrm{d}t}x_1 = ax_1, & x_1(t_0) = x_0 \\ \frac{\mathrm{d}}{\mathrm{d}t}x_2 = ax_2, & x_2(s) = g(s). \end{cases}$$

Theorem 4.12. For $n \geq 2$, the solution to $\frac{\mathrm{d}}{\mathrm{d}t}x = Ax + G(t)$, $x(t_0) = x_0$ is given by

$$\begin{aligned} x(t) &= e^{A(t-t_0)}x_0 + \int_{t_0}^{t} e^{A(t-s)}G(s)\,\mathrm{d}s \\ &= e^{A(t-t_0)}\left[x_0 + \int_{t_0}^{t} e^{-A(s-t_0)}G(s)\,\mathrm{d}s\right]. \end{aligned}$$

Proof. Define

$$c(t) := x_0 + \int_{t_0}^{t} e^{-A(s-t_0)}G(s)\,\mathrm{d}s.$$

Then the proposed solution is $x(t) = e^{A(t-t_0)}c(t)$. In light of existence and uniqueness, we just need to verify that this solves the ODE with the IC. We have

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[e^{A(t-t_0)}c(t)\right] = Ae^{A(t-t_0)}c(t) + e^{A(t-t_0)}\dot{c}(t).$$

Now, using the fact that $\frac{\mathrm{d}}{\mathrm{d}t}\left(\int_{t_0}^{t} b(s)\,\mathrm{d}s\right) = b(t)$, we have

$$\dot{c}(t) = e^{-A(s-t_0)}G(t)\Big|_{s=t} = e^{-A(t-t_0)}G(t),$$

Thus

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[e^{A(t-t_0)}c(t)\right] = Ae^{A(t-t_0)}c(t) + e^{A(t-t_0)}\dot{c}(t) = Ax + G(t).$$

□

Remark 4.13. From this result we have **Duhamel's formula**: The differential equation

$$\partial_t f = Lf + g(t,x), \qquad f|_{t=0} = f_0$$

where L is a t-independent linear operator (e.g., ∂_x), has solution

$$f = e^{Lt}f_0 + \int_0^t e^{L(t-s)}g(s,x)\,\mathrm{d}s.$$

5 The Theory of Existence and Uniqueness

We focus on the case $n = 1$, but the proof generalizes to $n \geq 2$ easily.

Theorem 5.1 (Existence and Uniqueness). Consider the differential equation

$$\frac{\mathrm{d}}{\mathrm{d}t} y = f(t, y(t)), \qquad y(t_0) = y_0$$

in the region $R : |t - t_0| \leq a$, $|y - y_0| \leq b$. Assume that f is continuous in R and Lipschitz in y with Lipschitz constant L. Then, there exists $h > 0$ such that the ODE admits a unique C^1 solution for $|t - t_0| \leq h$. Moreover, h can be chosen to any value satisfying $0 < h < \min\{a, b/M, 1/L\}$.

Proof *(Picaro iteration)***.** Define $\varphi_0 = y_0$, and for each $n \geq 0$, $\varphi_{n+1}(t) := y_0 + \int_{t_0}^{t} f(s, \varphi_n(s))\,\mathrm{d}s$. We will show first that $|\varphi_n(t) - y_0| \leq b$ in a small neighborhood of t_0. From f being continuous, we know that $|f| \leq M$ in R. Thus

$$|\varphi_{n+1}(t) - y_0| \leq \int_{t_0}^{t} |f(s, \varphi_n(s))|\,\mathrm{d}s \leq M|t - t_0|.$$

And so by setting $h \leq b/M$, we have $|\varphi_n(t) - y_0| \leq b$ for all n and $|t - t_0| \leq h$. Next, we show that $|\varphi_{n+1} - \varphi_n| \to 0$ as $n \to \infty$. Note that

$$|\varphi_{n+1}(t) - \varphi_n(t)| = \left| \int_{t_0}^{t} f(s, \varphi_n(s)) - f(s, \varphi_{n-1}(s))\,\mathrm{d}s \right| \leq L \int_{t_0}^{t} |\varphi_n(s) - \varphi_{n-1}(s)|\,\mathrm{d}s.$$

Note recall that we have the uniform in time bound $|\varphi_1 - \varphi_0| = |\varphi_1 - y_0| \leq Mh$. Iterating this inequality

$$|\varphi_{n+1}(t) - \varphi_n(t)| \leq (Lh)^n Mh$$

which converges if we choose h such that $Lh < 1$. We define thus

$$\varphi := \lim_{n\to\infty} ved\varphi_n = \lim_{n\to\infty} \sum_{i=1}^{\infty} (\varphi_{i+1} - \varphi_i) + \varphi_0.$$

Note that the series on the right converges uniformly in light of the bound above.

To see φ solves the ODE, note that from $\varphi(t) = y_0 + \int_{t_0}^{t} f(s, \varphi(s))\,\mathrm{d}s$, we have

$$\frac{\mathrm{d}}{\mathrm{d}t} \varphi(t) = f(t, \varphi(t)) \implies \varphi \in C^1.$$

It remains to show uniqueness. Suppose that ψ is another solution to the ODE with the same IC. Note that

$$\varphi(t) = t_0 + \int_{t_0}^{t} f(s, \varphi(s))\,\mathrm{d}s, \quad \psi(t) = t_0 + \int_{t_0}^{t} f(s, \psi(s))\,\mathrm{d}s.$$

Thus

$$d(t) := \varphi(t) - \psi(t) = \int_{t_0}^{t} [f(s, \varphi(s)) - f(s, \psi(s))]\,\mathrm{d}s \implies |d(t)| \leq L \int_{t_0}^{t} |d(s)|\,\mathrm{d}s.$$

Gronwall's inequality gives $d(t) \leq 0$ for $|t - t_0| \leq h$. □

Remark 5.2. The assumption $f \in C^0$ is sufficient for existence. The Lipschitz assumption is sufficient for uniqueness.

6 Quantitative Estimates

Lemma 6.1 (Gronwall's Inequality). For given α, β, if

- $\beta(t) \geq 0$,
- $\psi(t) \leq \alpha(t) + \int_0^t \beta(s)\psi(s)\,\mathrm{d}s$,
- $B(s) \coloneqq \exp\left(\int_0^s \beta(t)\,\mathrm{d}t\right)$,

then

$$\psi(t) \leq \alpha(t) + \int_0^t \alpha(s)\beta(s)\frac{B(t)}{B(s)}\,\mathrm{d}s.$$

If $\alpha \in C^1$, from integration by parts we have

$$\psi(t) \leq \alpha(0)B(t) + \int_0^t \alpha'(s)\frac{B(t)}{B(s)}\,\mathrm{d}s.$$

Proof. Write $A(t) \coloneqq \int_0^t \beta(s)\psi(s)\,\mathrm{d}s$. From $\psi(t) \leq \alpha(t) + A(t)$ and $\beta \geq 0$, we have

$$\frac{\mathrm{d}}{\mathrm{d}t}A(t) = \psi\beta \leq \alpha\beta + A\beta.$$

This is a first order linear "ODE inequality" in A. Using the integrating factor $1/B(t)$, we have

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{A(t)}{B(t)}\right) \leq \frac{\alpha(t)\beta(t)}{B(t)},$$

where we note $B^{-1} = \exp\left(-\int_0^t \beta\,\mathrm{d}t\right)$ and $\dot{B}^{-1} = -\beta B^{-1}$.

Integrating both sides from 0 to t gives

$$A(t)B^{-1}(t) \leq \int_0^t \alpha(s)\beta(s)B^{-1}(s)\,\mathrm{d}s + A(0)B^{-1}(0),$$

where the last term is zero since $A(0) = 0$ by definition. Now we have

$$\psi(t) \leq \alpha(t) + A(t) \leq \alpha(t) + \int_0^t \alpha(s)\beta(s)\frac{B(t)}{B(s)}\,\mathrm{d}s.$$

□

Remark 6.2. Applications:

- $\alpha = 0$ implies $\psi \leq 0$.
- $\alpha(t) = \alpha_0 + \alpha_1 t, \beta = L$. Then $B(s) = \exp\left(\int_0^s L\,\mathrm{d}s\right) = e^{Ls}$ and so

$$\psi(t) \leq \alpha_0 e^{Lt} + \alpha_1 \int_0^t e^{L(t-s)}\,\mathrm{d}s.$$

In some sense this is similar to the Duhamel's formula for the ODE system

$$\begin{cases} \frac{\mathrm{d}}{\mathrm{d}t}x = Lx + \alpha_1 \\ x(0) = \alpha_0 \end{cases} \implies x(t) = e^{Lt}\alpha_0 + \alpha_1 \int_0^t e^{L(t-s)}\,\mathrm{d}s.$$

Theorem 6.3 (Continuous Dependence). Suppose f is Lipschitz in the domain D of interest with Lipschitz constant L and denote

$$M := \max_{(t,x)\in D} |f(t,x) - g(t,x)|.$$

Then,

$$|x(t) - y(t)| \le e^{L|t-t_0|}|x_0 - y_0| + \frac{M}{L}\left(e^{L|t-t_0|} - 1\right),$$

where the first part is due to the IC error and the second part is due to the ODE error.

Proof. Note that

$$\frac{\mathrm{d}}{\mathrm{d}t}[x(t) - y(t)] = f(t, x(t)) - g(t, y(t)).$$

Integrating,

$$|x(t) - y(t)| \le |x_0 - y_0| + \int_{t_0}^t |f(s, x(s)) - g(s, y(s))|\,\mathrm{d}s.$$

The integrand may be bounded as

$$\begin{aligned} |f(s,x(s)) - g(s,y(s))| &\le |f(s,x(s)) - f(s,y(s))| + |f(s,y(s)) - g(s,y(s))| \\ &\le L|x(s) - y(s)| + M. \end{aligned}$$

Observe that

$$|x(t) - y(t)| \le |x_0 - y_0| + \int_{t_0}^t [L|x(s) - y(s)| + M]\,\mathrm{d}s.$$

Setting $\alpha := |x_0 - y_0| + M(t - t_0)$ and $\psi(t) := |x(t) - y(t)|$, we may apply Gronwall's inequality with $\beta = L$ to get

$$\begin{aligned} |x(t) - y(t)| &\le e^{L(t-t_0)}|x_0 - y_0| + \int_{t_0}^t Me^{L(t-s)}\,\mathrm{d}s \\ &\le e^{L|t-t_0|}|x_0 - y_0| + \frac{M}{L}\left(e^{L|t-t_0|} - 1\right). \end{aligned}$$

□

Corollary 6.4.

- If $f = g$, $x_0 = y_0$, then $x = y$.
- If $f = g$ then
$$|x(t) - y(t)| \leq e^{L|t-t_0|}|x_0 - y_0|.$$
That is, we have continuous (and in fact Lipschitz) dependence on the IC. In other words, if we define $\Phi(t, x_0)$ to be the solution of
$$\frac{\mathrm{d}}{\mathrm{d}t}\Phi = f(t, \Phi), \quad \Phi(t_0, x_0) = x_0,$$
then Φ is Lipschitz in x_0.
- If $x_0 = y_0$ and say $|f - g| \leq \varepsilon$, then
$$|x(t) - y(t)| \leq \frac{\varepsilon}{L}\left(e^{L|t-t_0|} - 1\right).$$
In particular, if $f_\alpha(t, x) \coloneqq \alpha f_0 + (1-\alpha) f_1$, then
$$|f_\alpha - f_\beta| \leq |\alpha - \beta| \left(|f_1| + |f_0|\right).$$

Now consider
$$y' \leq F(t, y(t)), \quad t \in [a, b].$$

Theorem 6.5 (Comparison). Suppose F is Lipschitz. Let $f, g \in C^1$ are such that
$$f' \leq F(t, f(t)), \quad g' = F(t, g(t)).$$
If $f(a) \leq g(a)$, then
$$f(t) \leq g(t), \quad t \in [a, b].$$

Proof. Suppose for contradiction that $f(t_1) > g(t_1)$. Let $\Omega \coloneqq \{t : f(t) \leq f(t)\}$. We have $a \in \Omega$. Let $t_0 \coloneqq \max\{t \in [a, t_0] \cap \Omega\}$. We know $f(t_0) = g(t_0)$ and $f(t) \leq g(t)$ for each $t \in \Omega$. Moreover, $f(t) > g(t)$ for each $t \in (t_0, t_1]$. But this and continuity implies that $f(t_0) \geq g(t_0)$. Thus $f(t_0) = g(t_0)$.

Now, from assumption we have
$$f'(t) - g'(t) \leq F(t, f(t)) - F(t, g(t))$$
By integrating both sides,
$$\begin{aligned} f(t) - g(t) &\leq f(t_0) - g(t_0) + \int_{t_0}^{t} F(s, f(s)) - F(s, g(s))\,\mathrm{d}s \\ &\leq \int_{t_0}^{t} L(f(s) - g(s))\,\mathrm{d}s, \end{aligned}$$

where the second inequality is justified from $f(t) \geq g(t)$ on $(t_0, t_1]$. By Gronwall's inequality, we have

$$f(t) - g(t) \leq 0, \quad t \in [t_0, t_1]$$

a contradiction. □

Now, if F is not Lipschitz, this no longer holds true:

Example 6.6.

$$\frac{\mathrm{d}}{\mathrm{d}t} y = y^{\frac{1}{3}}, \quad y(0) = 0$$

has solution $g(t) \equiv 0$. Now try the ansatz $f(t) = ct^{\alpha}$. We have

$$f'(t) = c\alpha t^{\alpha - 1}, \quad f^{\frac{1}{3}}(t) = c^{\frac{1}{3}} t^{\frac{\alpha}{3}},$$

giving

$$\alpha = \frac{3}{2}, \quad c = \pm \left(\frac{3}{2}\right)^{\frac{3}{2}}.$$

In particular, a solution is

$$f(t) = \begin{cases} ct^{\frac{3}{2}}, & t \geq 0 \\ -c(-t)^{\frac{3}{2}}, & t < 0 \end{cases},$$

which crosses g at $t = 0$ and is C^1 (since in particular both parts of f is C^1 at 0).

Example 6.7. Consider

$$\frac{\mathrm{d}}{\mathrm{d}y} x_{\alpha}(t) = f(t, x_{\alpha}(t)) + \alpha, \quad x_{\alpha}(t_0) = x_0.$$

Then the solution satisfies $x_{\alpha}(t) \leq x_{\beta}(t)$ if $\alpha \leq \beta$.

6.1 Extension

Lemma 6.8 (Gluing). Suppose $f \in C(\mathbb{R}, \mathbb{R})$ and $f(t, x)$ is Lipschitz in any bounded region. For $i = 1, 2$, suppose $\varphi_i(t)$ solves

$$\frac{\mathrm{d}}{\mathrm{d}t} x = f(t, x), \quad x(t_0) = x_0$$

on $t \in I_i := (a_i, b_i)$. If $t_0 \in I_1 \cap I_2 = (a, b)$ and $\varphi_1(t_0) = \varphi_2(t_0)$, then φ_1 and φ_2 agrees on $I_1 \cap I_2$ and

$$\varphi(t) := \begin{cases} \varphi_1(t), & t \in I_1 \\ \varphi_2(t), & t \in I_2 \end{cases}$$

is a C^1 solution to the ODE on $I_1 \cup I_2$.

We will show only that $\varphi_1 = \varphi_2$ on $I_1 \cap I_2$. The rest is easy.

Proof *(Method 1)***.** Apply existence and uniqueness to the ODE with IC at t_0 to get $\varphi_1 = \varphi_2$ on $t_0 \pm h$. Write

$$J_- := \{p : \varphi_1(t) = \varphi_2(t), t \in [p, t_0 + h] \cap (a, b)\}.$$

We will show that $a = \inf J_- =: A$. Suppose not, then $A > a$. There exists a sequence $P_n \to A$, each $P_n \in J_-$, so that $\varphi_1 = \varphi_2$ on each $[P_n, t_0 + h]$. In particular, $\varphi_1(P_n) = \varphi_2(P_n)$. Thus by continuity, $\varphi_1(A) = \varphi_2(A)$. But from existence and uniqueness we know $\varphi_1 = \varphi_2$ on $[A - h', t_0 + h]$ for some $h' > 0$, contradicting the definition of A. □

Proof *(Method 2)***.** Set $J := \{t \in I_1 \cap I_2 : \varphi_1(t) = \varphi_2(t)\}$. The goal is to show $J = I_1 \cap I_2, J \neq \varnothing$. By continuity, J is relatively closed. But since it is also relatively open by local existence and uniqueness, we have $J = I_1 \cap I_2$. □

Proof *(Method 3)***.** Using Gronwall's inequality, from

$$|\varphi_1(t) - \varphi_2(t)| \leq \int_{t_0}^{t} L|\varphi_1(s) - \varphi_2(s)| \, \mathrm{d}s,$$

we have $\varphi_1 = \varphi_2$ on $I_1 \cap I_2$. □

7 Blowup

Theorem 7.1 (Blowup Criterion). Suppose $f \in C(\mathbb{R}, \mathbb{R})$ is Lipschitz in x in any bounded region. Suppose that φ is a C^1 solution to $\frac{\mathrm{d}}{\mathrm{d}t}x = f(t, x)$ on (t_-, t_+). The solution can be extended to $(t_-, t_+ + \varepsilon)$ for some $\varepsilon > 0$ if and only if one of the following equialent conditions hold:

(i) There exists a sequence $\{t_n\}$ such that $t_n \to (t_+)^-$ and $\varphi(t_n) \to \varphi_0 \neq \pm\infty$. Note that this is true for oscillating behavior but not monotonic blowup.

(ii) $\liminf_{t \to (t_+)^-} |\varphi(t)| < +\infty$.

In particular, the solution cannot be extended beyond t_+ if $\lim_{t \to (t_+)^-} |\varphi(t)| = +\infty$.

Proof. The forward direction is straightforward. For the converse, suppose $|\varphi(t_n) - \varphi_0| \leq 1$ for each $n \geq N$. The idea is to apply existence and uniqueness at $(t_n, \varphi(t_n))$ for large n and hope that we get a solution on $(t_n - h, t_n + h)$ with $t_n + h > t_+$.

Write

$$S \coloneqq [t_+ - 1, t_+ + 1] \times [\varphi_0 - 1, \varphi_0 + 1],$$
$$D \coloneqq [t_+ - 2, t_+ + 2] \times [\varphi_0 - 2, \varphi_0 + 2].$$

We will control $(t_n, \varphi(t_n))$ to be in S, and φ to be in D. Since f is continuous and Lipschitz in D, we have

$$|f| \leq M, \quad |f(t, x) - f(t, y)| \leq L|x - y|$$

in D for some $L, M > 0$. In particular, $|\varphi'| \leq M$ and a solution starting from S cannot escape from D in any time less than $1/M$. Applying existence and uniqueness at $(t_n, \varphi(t_n))$ for large n gives a unique solution ψ such that $\varphi(t_n) = \varphi(t_n)$ with domain $(t_n - h, t_n + h)$ where h can be chosen to be anything such that

$$0 < h < \min\left\{1, \frac{1}{M}, \frac{1}{L}\right\}.$$

By choosing large n, we have $t_n + h > t_+$. □

Proposition 7.2. Consider $x' = f(t, x)$ with $|f| \leq M$ for each (t, x), $f \in C(\mathbb{R}, \mathbb{R})$ and Lipschitz in x in any bounded region. Then the solution exists globally and is unique.

Proof. Assume the solution exists on (t_-, t_+). Note that

$$|x(t) - x(t_0)| = \left| \int_{t_0}^{t} f(s, x(s))\, \mathrm{d}s \right| \leq M|t - t_0| < \infty.$$

By the blowup criterion, we know that x can be extended beyond t_+ and t_-. □

Example 7.3 (Riccatti ODE). Consider the Riccatti ODE $y' = y^p$, $y(0) = 1$.

- $p = 1$. The unique solution is $y(t) = e^t$.
- $p > 1$. By the comparison principle we have $y \geq e^t$ since $y(0) \geq 1$. By separation of variables, we have $y'/y^p = 1$ and then

$$y^{p-1}(t) = \frac{1}{1-(p-1)t} \to +\infty \text{ as } t \to \left(\frac{1}{p-1}\right)^- .$$

  So we have blow up at $t_+ = 1/(p-1) < \infty$.

Example 7.4. Consider $y' = e^y \geq y^2$. By comparison with $y' = y^2$, we have finite time blowup. Consider $y' = y^n + \cdots + 1 + e^t$, $y(0) = 1$. By comparison with $y' = y^2$, we have finite time blowup.

Example 7.5. $y' = y^2$, $y(0) = y_0$.

- If $y_0 > 0$, we have finite time blowup.
- If $y_0 = 0$, we have $y \equiv 0$.
- If $y_0 < 0$, we have $y = y_0/(1 - ty_0)$. Note that $1 - ty_0 > 0$ for $t > 0$. Thus $y_0 \sim -1/t$ as $t \to +\infty$.

From this we see nonlinear stabilization: negative IC gives global solution decaying to 0, while positive IC gives finite time blowup.

Example 7.6. Consider $y' = y^2 + \varepsilon$ for $\varepsilon > 0$ and $y(0) = y_0$. Note that $y(t) \geq y_0 + \varepsilon t$ and so $y(t_0) > 0$ for t_0 large. Then, we have $\dot{y} > y^2$ and thus blowup.

Example 7.7. Consider $y' = y^2 - \varepsilon^2$. We have $y_1 \equiv \varepsilon$ and $y_2 \equiv -\varepsilon$. For $y_0 > \varepsilon$ we again have $y' \geq y^2$ and so blowup.

8 Autonomous ODE and Phase Portrait/Plane

Recall that a general first order ODE is $x' = F(t, x)$, where $x, F \in \mathbb{R}^{n\times 1}$. An **autonomous** ODE is one where F does not depend on t:

$$x' = F(x).$$

A special solution is the constant solution $x(t) \equiv x_*$. Such a solution is called a **equilibrium** or a **critical point**. Note that non autonomous ODEs in general does not has critical points, since F changes with t.

We call $(t, x(t))$ the **trajectory** of the solution $x(t)$. For $n = 2$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} f_1(x_1, x_2) \\ f_2(x_1, x_2) \end{pmatrix},$$

we call the x_1-x_2 plane the **phase plane**. We may think of x_2 as a function of x_1, $x_2 = x_2(x_1)$, to remove the dependency on t: By the chain rule,

$$\frac{\mathrm{d}x_2}{\mathrm{d}x_1} = \frac{\mathrm{d}x_2/\mathrm{d}t}{\mathrm{d}x_1/\mathrm{d}t} = \frac{f_1}{f_2}(x_1, x_2).$$

The representative set of trajectories is called the **phase portrait**.

8.1 Phase Portrait of 2 x 2 Systems

Consider the linear 2×2 system:

$$x' = Ax, \quad A \in \mathbb{R}^{2\times 2}.$$

The equilibrium is at $x_* = 0$ and the real eigenvectors associated with the eigenvalue 0. But what happens as $t \to \infty$ for a general solution x?

Remark 8.1. Recall first that for

$$z(t) := e^{\alpha t}\left(\cos\beta t + \mathrm{i}\sin\beta t\right),$$

we have

(i) If $\alpha > 0$, then $|z(t)| \to \infty$ as $t \to \infty$.

(ii) If $\alpha = 0$, then $z(t)$ is bounded, periodic, and in particular does not converge.

(iii) If $\alpha < 0$, then $z(t) \to 0$ exponentially fast as $t \to \infty$.

We consider the following three cases:

Case 1:

$\lambda_1 < \lambda_2$ real. The general solution is

$$x(t) = \underbrace{e^{\lambda_t}}_{\text{amplitude}} \underbrace{\left(e^{(\lambda_1-\lambda_2)t} c_1 v_1 + c_2 v_2\right)}_{\text{direction}}.$$

(i) If $\lambda_1 < 0$, then any solution of the form

$$x(t) = c_1 e^{\lambda_1 t} v_1$$

will converge to 0 as $t \to \infty$, remaining on the line spanned by v_1 on the way.

(ii) (**Sink**) $\lambda_1 < \lambda_2 < 0$, then $x(t) \approx e^{\lambda_2 t} c_2 v_2$ for large t. Thus $x(t) \to 0$ as $t \to \infty$ (converging in the v_1 direction faster).

(iii) (**Saddle**) $\lambda_1 < 0 < \lambda_2$. We have $x \to 0$ only if $x \equiv 0$.

(iv) (**Source**) $0 < \lambda_1 < \lambda_2$, then $x(t) \to \infty$ (diverging in the v_2 direction faster) as $t \to \infty$ unless $x \equiv 0$.

These can be visualized in a phase portrait.

Case 2:

$\lambda = \alpha \pm \mathrm{i}\beta$, $v = a \pm \mathrm{i}b$, $b \neq 0$. The general solution is

$$\begin{aligned} x(t) &= c_1 \operatorname{Re}(e^{\lambda t} v) + c_2 \operatorname{Im}(e^{\lambda t} v) \\ &= e^{\alpha t} \left[c_1 (\cos \beta t a - \sin \beta t b) + c_2 (\sin \beta t a + \cos \beta t b)\right] \\ &= e^{\alpha t} V(t), \end{aligned}$$

where

$$V(t) := c_1 (\cos \beta t a - \sin \beta t b) + c_2 (\sin \beta t a + \cos \beta t b)$$

is periodic in t.

(i) If $\alpha < 0$, $|x| \to 0$.

(ii) If $\alpha = 0$, then $x(t) = V(t)$ is bounded and periodic (but does not converge).

(iii) If $\alpha > 0$, then $|x| \to \infty$.

Case 3:

One real eigenvalue with two linearly independent eigenvectors. General solution

$$x(t) = e^{\lambda t} \underbrace{(c_1 v_1 + c_2 (t v_1 + v_2))}_{\text{constant}}.$$

The trajectory is along a straight line.

Case 4:

One eigenvector v_1 and one generalized eigenvector v_2 such that $(A-\lambda)^2 v_2 = 0$. General solution:

$$\begin{aligned} x(t) &= e^{\lambda t}\Big(c_1 v_1 + c_2 t v_2\Big) \\ &= e^{\lambda t} t \left(\frac{1}{t} c_1 v_1 + c_2 v_2\right). \end{aligned}$$

For large t, the trajectory is approximately along the line spanned by v_2.

(i) If $\lambda > 0$, $|x| \to \infty$ exponentially fast.

(ii) If $\lambda = 0$, $|x| \sim t$.

(iii) $\lambda < 0$, $|x| \to 0$ exponentially fast.

Q: How smooth is x_2 as a function of x_1? Obviously Case 3 is C^∞.
Nonlinear phase plane/portrait is much harder:

Example 8.2.

$$\begin{cases} x_1' = f_1(x_1, x_2) \\ x_2' = f_2(x_1, x_2) \end{cases}.$$

Suppose for example that f_i are polynomials. We may have a few critical points, say $(0,0)$ and x_c. A hard problem: find a C^∞ trajectory starting from 0 and passing x_c.

Note that by existence and uniqueness, trajectories cannot cross (except at critical points).

8.2 Stability

We say the critical point x_* is

- **stable** if for any $\varepsilon > 0$, there exists $\delta > 0$ such that if $|x_0 - x_*| < \delta$, $|x(t) - x_*| < \varepsilon$ for each $t \geq t_0$.
- **asymptotically stable** if there exists $\delta > 0$ such that if $|x_0 - x_*| < \delta$, then $\lim_{t\to\infty} x(t) = x_*$.
- **unstable** if it is not stable.

We summarize the stability of the critical point $(0,0)$ for the 2×2 linear systems:

(i) If $\operatorname{Re}\lambda_i < 0$ for $i = 1, 2$, then both stable and asymptotically stable (sink).

(ii) If $\operatorname{Re}\lambda_i > 0$ for some i, then unstable.

(iii) If $\operatorname{Re}\lambda_i \leq 0$ for all i and $\operatorname{Re}\lambda_i = 0$ for some i, then not asmptotically stable. Moreover,

(i) if $\lambda_1 \neq \lambda_2$, then stable ($|e^{\lambda_1 t}| = |e^{\lambda_2 t}| \leq 1$).

(ii) if $0 = \lambda_1 = \lambda_2$, 2 eigenvectors, stable.

(iii) $0 = \lambda_1 = \lambda_2$, where λ_2 is the generalized eigenvector, then unstable (since $|x(t)| \sim t$).

Example 8.3. Consider the pendulum equation

$$\theta'' + \gamma\theta' + w^2 \sin\theta = 0.$$

It has two critical points: $(0, 0)$ and $(\pi, 0)$. The latter corresponds to the inverted pendulum and is unstable.

Lemma 8.4. For each $\varepsilon > 0$ and integer $n \geq 0$, there exists a constant $C > 0$ such that

$$t^n \leq Ce^{\varepsilon t}, \quad t \geq 0.$$

Proof. Note that

$$t^n = e^{\frac{n}{\varepsilon} \cdot \varepsilon \log t}, \quad Ce^{\varepsilon t} = e^{\varepsilon C + \varepsilon t}.$$

The first exponent is concave. □

Theorem 8.5 (Linear Stability). Consider the ODE system $x' = Ax$, $A \in \mathbb{R}^{n\times n}$, with $x_* = 0$ as a critical point. Let λ_i be the eigenvalues of A.

(i) If $\operatorname{Re}\lambda_i < 0$ for each i, then x_* is stable and asymptotically stable.

(ii) If $\operatorname{Re}(\lambda_i) > 0$ for some i, then x_* is unstable.

(iii) If $\operatorname{Re}(\lambda_i) \geq 0$ for some i, then x_* is not asymptotically stable.

(iv) If $\operatorname{Re}(\lambda_i) \leq 0$ for each i and if the Jordan block J_i associated with λ_i, $\operatorname{Re}(\lambda_i) = 0$, is 1×1, then x_* is stable. If the block is larger than 1×1, then x_* is unstable.

Remark 8.6. To gain some intuition of why only the real parts matter, note that

$$|e^{(\alpha+\mathrm{i}\beta)t}| = |e^{\alpha t}(\cos\beta t + \mathrm{i}\sin\beta t)| = e^{\alpha t}.$$

If $\alpha > 0$, $|x(t)| \to \infty$ and x_* is unstable. If $\alpha = 0$, $|x(t)| \equiv |x_0|$ and x_* is stable. If $\alpha < 0$, $|x(t)| \to 0$ and x_* is asymptotically stable.

Proof. Recall that the solution is $x(t) = e^{At}x_0$. Write $A = UJU^{-1}$ and note that

$$e^{At} = Ue^{Jt}U^{-1}, \quad e^{Jt} = \begin{pmatrix} e^{J_1 t} & & 0 \\ & \ddots & \\ 0 & & e^{J_k t} \end{pmatrix},$$

where J_j are the Jordan blocks of A.

We start from case (ii). Note that $x(t) = e^{\lambda_i} v_i$ solves the ODE (if λ_i is complex, take the real or complex parts of x). We have

$$|x(t)| = |cv_i| e^{\mathrm{Re}(\lambda_i)t} \longrightarrow \infty.$$

Thus x_* is unstable.

Next, consider case (iii). If $|\lambda_i| \geq 0$, we have with a small perturbation of size c that

$$|x(t)| = |cv_i| e^{\mathrm{Re}(\lambda_i)t} \geq |cv_i|,$$

which does not converge to 0.

Now consider (i). Recall that

$$e^{J_j t} = \begin{pmatrix} e^{\lambda_j t} & te^{\lambda_j t} & \frac{t^2}{2} e^{\lambda_j t} & \cdots & \frac{t^{m_j-1}}{(m_j-1)!} e^{\lambda_j t} \\ 0 & e^{\lambda_j t} & te^{\lambda_j t} & \cdots & \frac{t^{m_j-2}}{(m_j-2)!} e^{\lambda_j t} \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & 0 & \cdots & e^{\lambda_j t} \end{pmatrix}.$$

We claim (Ex.) that $|t^m| \leq e^{\varepsilon t} C_{m,\varepsilon}$ for each $t \geq 0$. (E.g., $t^{1000} \leq e^{0.1t} C$ for each $t \geq 0$.) We have the estimate

$$|e^{\lambda_i t} t^m| \leq |e^{\mathrm{Re}\,\lambda_i t} t^m| \leq C_A e^{-(2/3)\lambda t},$$

where we pick $\varepsilon = \lambda/3$. This gives

$$|(e^{J_j t})_{kl}| \leq C_A e^{-(2/3)\lambda t} \implies |(e^{Jt})_{ij}| \leq C_A e^{-(2/3)\lambda t}.$$

Now note that

$$|(e^{At})_{ij}| \leq \sum_{k,l} |U_{ik}| |(e^{Jt})_{kl}| |(U^{-1})_{lj}|.$$

We thus have

$$|(e^{At})_{ij}| \leq C_a n^2 C_a e^{-(2/3)\lambda t} \leq C_A e^{-(2/3)\lambda t}.$$

Then,

$$|e^{At} x_0| \leq C_A e^{-(2/3)\lambda t} |x_0| \to 0.$$

Finally, consider (iv). Recall that

$$e^{J_i t} = \begin{pmatrix} e^{\lambda_i t} & te^{\lambda_i t} & \cdots & \frac{t^{n_i-1}}{(n_i-1)!} e^{\lambda_i t} \\ 0 & e^{\lambda_i t} & \cdots & \frac{t^{n_i-2}}{(n_i-2)!} e^{\lambda_i t} \\ \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & e^{\lambda_i t} \end{pmatrix}.$$

If $n_i \geq 2$ then x_* is unstable. Ex: find x_0 cleverly such that $e^{At} x_0 \to \infty$. If $n_i = 1$ for each n_i, then $|e^{J_i t}| \approx 1$. □

Example 8.7. $x' = \begin{pmatrix} -2 & \\ & -1 \end{pmatrix} x$ is asymptotically stable and stable. $x' = \begin{pmatrix} & 1 \\ -1 & \end{pmatrix} x$ has eigenvalues $\pm \mathrm{i}$ and is stable but not asymptotically stable.

8.3 Nonlinear Stability

Consider

$$x' = F(x), \quad F(x_*) = 0, \quad F \in C^2.$$

We linearize around x_*:

$$x' = F(x) = F(x_*) + \nabla F(x_*)(x - x_*) + G(x - x_*).$$

If $|x - x_*| < 1$, we have $G(x - x_*) \le M|x - x_*|^2$ for some $M > 0$. Write $z(t) = x(t) - x_*$ and note that

$$z'(t) = x'(t) = Az + G(z), \quad A := \nabla F(x_*),$$

where

$$|G(z)| \le M|z|^2, \quad \forall |z| < 1.$$

This system has the equilibrium $z_* = 0$. The hope is that for small z, $|G(z)| \le M|z|^2 \ll |z|$, and the solution of the nonlinear system behaves like that of the linear system $z' = Az$.

Theorem 8.8 (Nonlinear Stability). Let λ_i be the eigenvalues of $A := \nabla F(x_*)$, where $F \in C^2$.

(i) If $\mathrm{Re}(\lambda_i) < 0$ for each i, then x_* is asymptotic stable and stable.

(ii) If for some i we have $\mathrm{Re}\,\lambda_i > 0$, then x_* is unstable.

In other cases (say $\mathrm{Re}\,\lambda_i = 0$) it is unclear (we need some information from G.)

Example 8.9. Consider $x' = 0 + x^2$. The IC $\varepsilon > 0$ results in finite time blowup, while IC $\varepsilon < 0$ results in global solution converging to 0.

Proof. Case (ii) is left as an exercise. Consider (i).

Idea: $z' = Az + o(1)Z$. Suppose $\delta = |z_0| \ll 1$ and note that

$$|z(t)| \approx |e^{At} z_0| \le C_A e^{-(2/3)\lambda t} |z_0|$$

and

$$|G(z(t))| \le M|z(t)|^2 \le M C_A^2 \delta^2 e^{-(4/3)\lambda t}.$$

Goal: If $|z_0| = \delta \ll 1$, then $d(t) := e^{\frac{3}{2}\lambda t}|z(t)| \le \tilde{C}_A \delta$.

Step 1: the solution exists for a short time. Step 2: By Duhamel formula,

$$z(t) = \overbrace{e^{At}}^{I} z_0 + \overbrace{\int_0^t e^{A(t-s)} \underbrace{G(z(s))}_{F(s)} \, \mathrm{d}s}^{II}.$$

We hope to derive an ODE inequality for $d(t)$. Denote as I and II the two terms on the RHS. We have

$$|I| \leq C_A e^{-\frac{2}{3}\lambda t}|z_0|$$

and

$$|II| \leq C_A \int e^{-\frac{2}{3}\lambda(t-s)}|G(z(s))| \leq \tilde{C}_A \int e^{-\frac{2}{3}\lambda(t-s)}|z(s)|^2.$$

Since $|z(s)| = d(s) \cdot e^{-\frac{2}{3}\lambda t}$, we have

$$|II| \leq \tilde{C}_A e^{-\frac{2}{3}\lambda t} \int_0^t \underbrace{e^{\frac{2}{3}\lambda s} e^{-\frac{4}{3}\lambda s} d(s)^2}_{|z(s)|^2} \, \mathrm{d}s.$$

This gives

$$d(t) = e^{\frac{2}{3}\lambda t}|z(t)| \leq e^{\frac{2}{3}\lambda t}\,[|I| + |II|] \leq C_A\delta + \tilde{C}_A \int_0^t e^{-\frac{2}{3}\lambda s} d(s)^2 \, \mathrm{d}s.$$

We now claim that $d(t) < 2C_A\delta$ for each $t \geq 0$ if $\delta \ll 1$. We induct on time (this is called a bootstrap argument). Base case $t = 0$: $d(0) = |z_0| = \delta$. Assume that the claim is true on $[0, t)$; from this we will prove that it remains true at time t:

$$\begin{aligned} d(t) &\leq C_A\delta + \tilde{C}_A \int_0^t e^{-\frac{2}{3}\lambda s} d^2(s) \, \mathrm{d}s \\ &\leq C_A\delta + \tilde{C}_A (2C_A)^2 \int_0^t e^{-\frac{2}{3}\lambda s}\delta^2 \, \mathrm{d}s \\ &\leq C_A\delta + C_{A,2}\delta^2. \end{aligned}$$

Take $\delta \leq C_A/2/C_{A,2}$ and we have

$$d(t) \leq C_A\delta + \frac{1}{2}C_A\delta < 2C_A\delta.$$

Equivalently,

$$|z(t)| < 2C_A\delta e^{-\frac{2}{3}\lambda t} \longrightarrow 0.$$

□

Example 8.10. Consider the pendulum system

$$m\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + \frac{mg}{L}\sin\theta = 0 \sin\theta = 0, \quad x = \theta, \quad y = \theta'.$$

This can be summarized as

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} y \\ -\frac{g}{L}\sin x \end{pmatrix}.$$

The critical points are $(n\pi, 0)$, $n \in \mathbb{Z}$.

Note that

$$\nabla F(x_*) = \begin{pmatrix} 0 & 1 \\ -\frac{g}{L}\cos(n\pi) & 0 \end{pmatrix}$$

If n is odd, we have

$$\nabla F(x_*) = \begin{pmatrix} 0 & 1 \\ \frac{g}{L} & 0 \end{pmatrix},$$

which has eigenvalues $\pm\sqrt{g/L}$. This corresponds to the inverted pendulum and is unstable.

If n is even, we have

$$\nabla F(x_*) = \begin{pmatrix} 0 & 1 \\ -\frac{g}{L} & 0 \end{pmatrix},$$

which has eigenvalues $\pm\mathrm{i}\sqrt{g/L}$. This corresponds to the hanging pendulum. We cannot conclude stability from linearization.

8.4 Liapunov

Note that the linearized method works only for $|x_0 - x_*| \ll 1$, since we are using Taylor expansion.

The Liapunov method gives a way to prove stability for larger perturbations. Accordingly, it is at times called the global method. However, while linearization applies for almost any autonomous system, the Liapunov method requires very special functions.

Definition 8.11. Consider a generalization of distance, a C^1 function $L : D \subset \mathbb{R}^n \to \mathbb{R}$. Here, D is open.

Example 8.12. $L(X) = \|x - x_*\|^2 = \sum (x_i - x_{*,i})^2$.

We now consider the evolution of $L(x(t))$:

$$\frac{\mathrm{d}}{\mathrm{d}t}L(x(t)) = \frac{\mathrm{d}}{\mathrm{d}t}x(t) \cdot \nabla L(x(t)) = F(x(t)) \cdot \nabla L(x(t)).$$

Write $\dot{L}(x) = F \cdot \nabla L|_x$ and we have

$$\frac{\mathrm{d}}{\mathrm{d}t} L(x(t)) = \dot{L}(x(t)).$$

Theorem 8.13 (Liapunov Stability). Let x_* be an equilibrium of the autonomous system $x' = F(x)$, $F \in C^1$. Assume that there exists L such that $L(x_*) = 0$ and $L(x) > 0$ for each $x \neq x_*$.

Now, if in addition, $\dot{L}(x) \leq 0$ for each $x \in D \setminus \{x_*\}$, then x_* is stable. If $\dot{L}(x) < 0$ for each $x \neq x_*$, then x_* is asymptotically stable.

The idea is that L is like a generalized distance from x_* and $\dot{L} \leq 0$ means the distance is non-increasing along trajectories.

The potential energy of the pendulum is $U(x, y) = mgh = mgL(1 - \cos x)$, where L is the length of the pendulum. (Recall that $x = \theta$ and $y = \theta'$.)

The kinetic energy is $\frac{1}{2}mv^2 = \frac{1}{2}mL^2y^2$. (Note that $v = L\dot{\theta} = Ly$.)

Define the **energy** E of the pendulum as

$$L(x, y) = \frac{1}{2}mL^2y^2 + mgL(1 - \cos x).$$

We claim that L is a Liapunov function for the hanging pendulum at $(0, 0)$:

- $L(0, 0) = 0$ and $L(x, y) > 0$ for each $(x, y) \neq (0, 0)$ (when we restrict the domain to $D := (-\pi, \pi) \times \mathbb{R}$).
- $\dot{L} = \nabla L \cdot F$.
$$\nabla L = \begin{pmatrix} mgL \sin x \\ mL^2 y \end{pmatrix}, \quad F = \begin{pmatrix} y \\ -\frac{g}{L} \sin x \end{pmatrix}.$$
So,
$$\dot{L} = \begin{pmatrix} mgL \sin x \\ mL^2 y \end{pmatrix} \cdot \begin{pmatrix} y \\ -\frac{g}{L} \sin x \end{pmatrix} = mgLy \sin x - mgLy \sin x = 0.$$
Thus $L(x(t)) \equiv L(x_0)$ does not change along trajectories. Thus $(0, 0)$ is not asymptotically stable.

Proof *(of Liapunov Stability Theorem)***.** Recall first that $\frac{\mathrm{d}}{\mathrm{d}t} L(x(t)) = \dot{L}(t)$.

Suppose $\dot{L}(x) \leq 0$ for each $x \neq x_*$. We hope to show that for each $\varepsilon > 0$, there exists $\delta > 0$ such that if $|x_0 - x_*| < \delta$, then $|x(t) - x_*| < \varepsilon$ for each $t \geq t_0$.

Without loss, suppose $B(x_*, \delta) \subset D$. Let $\alpha := \min\{L(y) : |y - x_*| = \delta\} > 0$. By continuity of L we pick $\varepsilon < \delta$ small so for each $|x_0 - x_*| < \varepsilon$ we have

$$0 \leq L(x_0) = L(x_0) - L(x_*) < \alpha.$$

Note that we have $\frac{\mathrm{d}}{\mathrm{d}t} L(x(t)) \leq 0$, so $L(x(t)) \leq L(x_0) < \alpha$. We claim that $x(t) \in B(x_*, \delta)$, since otherwise we have $|x(t) - x_0| = \delta$ for some t and then $L(x(t)) = \alpha$, a contradiction.

Now, suppose $\dot{L}(x) < 0$ for each $x \neq x_*$. Since this is a stronger condition, the above proof still works and shows that $x(t) \in B(x_*, \delta)$ for the same choice of ε. Suppose $\lim_{t\to\infty} x(t)$ does not converge to x_*. Then, $x(t_n) \to z_* \neq x_*$ for a sequence $t_n \to \infty$. But each time we visit z_*, the value $L(x(t))$ drops; after sufficiently many iterations, $L(x)$ must converge:

Note that for each $t \geq 0$, there exists large m such that

$$L(x(t)) \geq L(x(t_m)) \geq L(z_*) > 0,$$

where the second inequality comes from continuity. Consider the ODEs $z' = F(z)$, $z_0 = z_*$ and $y' = f(y)$, $y_0 = x(t_n)$. A solution to the second ODE is $y(0) = x(t_n)$, $y(1) = x(t_n + 1)$. We have $L(z_*) - L(z(1)) = \Delta > 0$. Note that

$$\begin{aligned} L(x(t_n + 1)) - L(z_*) &= L(x(t_n + 1)) - L(z_1) + L(z_1) - L(z_*) \\ &= L(y(1)) - L(z(1)) - \Delta \\ &\leq C|y(1) - z(1)| - \Delta, \end{aligned}$$

by f being Lipschitz. By continuous dependence the first term can be made less than $\Delta/2$ for large n, and we get a contradiction. □

Example 8.14. The ODE system

$$\begin{aligned} x' &= -x^3 \\ y' &= -y(x^2 + z^2 + 1) \\ z' &= -\sin z \end{aligned}$$

has equilibrium $x = y = z = 0$. Define $L(v) = x^2 + y^2 + z^2$.

$$\begin{aligned} \nabla L \cdot F &= 2x(-x^3) + 2y(-y(x^2 + z^2 + 1)) + 2z(-\sin z) \\ &= -2x^4 - 2y^2(x^2 + z^2 + 1) - 2z \sin z \leq 0. \end{aligned}$$

Take $D := \mathbb{R} \times \mathbb{R} \times (-\pi, \pi)$ and we have $\nabla L \cdot F < 0$ for each $(x, y, z) \neq (0, 0, 0)$.

Example 8.15.

$$v' = \begin{pmatrix} (\varepsilon x + 2y)(z+1) \\ (-x + \varepsilon y)(z+1) \\ -z^3 \end{pmatrix} =: F(v), \quad v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

Note that $v_* = (0,0,0)$ is the only critical point. We have

$$A = \nabla F(x_*) = \begin{pmatrix} \varepsilon & 2 & 0 \\ -1 & \varepsilon & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The eigenvalues of A satisfy $\lambda\left[(\lambda-\varepsilon)^2 + 2\right] = 0$, so $\lambda = 0, \varepsilon \pm \mathrm{i}\sqrt{2}$. Thus for $\varepsilon > 0$ the critical point is unstable. For $\varepsilon \le 0$, we use Liapunov method.

Let's try $L = ax^2 + by^2 + cz^2$. We have $L(v_*) = 0$ and $L(v) > 0$ for each $v \neq v_*$ if $a, b, c > 0$. Next, note that

$$\begin{aligned} \dot{L} = \nabla L \cdot F &= \begin{pmatrix} 2ax \\ 2by \\ 2cz \end{pmatrix} \begin{pmatrix} (\varepsilon x + 2y)(z+1) \\ (-x + \varepsilon y)(z+1) \\ -z^3 \end{pmatrix} \\ &= (z+1)\left[2ax(\varepsilon x + 2y) + 2by(-x + \varepsilon y)\right] - 2cz^4. \end{aligned}$$

The last term is always non-positive. The first term is

$$(z+1)\left[2a\varepsilon x^2 + 2b\varepsilon y^2 + (4a - 2b)xy\right],$$

where the first two terms in the bracket are non-positive if $\varepsilon \le 0$. Thus if $4a = 2b = 0$ and $z + 1 > 0$ we have on $D := \mathbb{R} \times \mathbb{R} \times (-1, \infty)$ the estimate $\dot{L} \le 0$. So the critical point is stable. If in addition $c > 0$, then $\dot{L} < 0$ for each $v \neq v_*$ and the critical point is asymptotically stable.

We next find the region of attraction. Let's fix say $\delta = 1$ and pick $L = x^2 + 2y^2 + z^2$. Note that $B(0,1) \subset D$. Recall in the proof of Liapunov stability, we defined

$$\alpha = \min_{\partial B(0,1)} L(x) = 1.$$

We want to find a ball $B(0,r) \subset \{x : L(x) < \alpha\}$. Note that

$$L(x) = x^2 + 2y^2 + z^2 \le 2(x^2 + y^2 + z^2) = 2|x|^2.$$

So if $|x| < 1/\sqrt{2}$, then $L(x) < 1 = \alpha$. Thus the region of attraction contains $B(0, 1/\sqrt{2})$.

Example 8.16. Consider the restriction

$$L = ax^2 + bxy + cy^2 > 0, \quad \forall (x, y) \neq (0, 0).$$

By considering $(x, 0)$ and $(0, y)$ we know that a, c must be positive. For the cross term, we can require

(i) $4ac > b^2$, or

(ii)
$$a\left(x + \frac{b}{2x}y\right)^2 + \left(c - \frac{b^2}{4a}\right)y^2 > 0, \quad \forall (x, y) \neq (0, 0).$$

This again requires $c - \frac{b^2}{4a} > 0$.

Example 8.17.
$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -x - xy^2 \\ -y - x^2y + Mx \end{pmatrix}.$$

Linearize at $(0, 0)$:
$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ M & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + O(|x|^2).$$

This gives $\lambda = -1$ and thus asymptotically stability.

For Liapunov, try $L = x^2 + cy^2$. Note that we must have $c > 0$.

$$\dot{L} = \nabla L \cdot F = \begin{pmatrix} 2x \\ 2cy \end{pmatrix} \cdot \begin{pmatrix} -x - xy^2 \\ -y - x^2y + Mx \end{pmatrix} = -2x^2 - 2cy^2 - 2x^2y^2 - 2cx^2y^2 + 2cMxy.$$

Dividing by -2, we hope to have

$$x^2 + cy^2 - cMxy > 0, \quad \forall (x, y) \neq (0, 0).$$

This is satisfied if $4c - c^2M^2 > 0$, or equivalently $c < \frac{4}{M^2}$.

Note that typically we have

$$\lim_{R \to \infty} L(\partial B_R) = \infty.$$

Thus we can pick large δ and large α to get a large region of attraction.

Remark 8.18. Typically we try polynomial Liapunov functions. But finding such functions is nontrivial when the system is complicated.

9 Periodic Solutions, The Poincare-Bendixson Theorem

This section concerns the 2×2 autonomous system $x' = F(x)$ where F is locally Lipschitz in x.

Note the following:

(i) If $x(x_0 + T) = x(t_0)$ for some $t_0, T > 0$, then by existence and uniqueness, $x(s + t) = x(s)$ for each $s \in \mathbb{R}$. That is x is periodic with period T.

(ii) x is a closed trajectory/orbit, that is, $\{x(t) : t \geq 0\}$ is a closed curve, if and only if it is a periodic solution.

Note that it is harder for trajectories to be periodic in higher dimensions, since it is less likely for trajectories to intersect. We will focus on $n = 2$.

Example 9.1. For $x' = Ax$ with $A \in \mathbb{R}^{2\times 2}$, if $\operatorname{Re}\lambda = 0$, $\lambda = \pm i\beta$, then

$$x(t) = \operatorname{Re}\left[e^{i\beta t} v_1 + e^{-i\beta t} v_2\right].$$

Thus every trajectory/orbit is periodic.

9.1 Limits Sets and Invariant Sets

We adopt the following definitions:

Definition 9.2.

(i) Use $\varphi_t(x)$ to denote the solution with IC x at time 0.

(ii) Write the trajectories forward and backward in time as

$$\gamma_+(x) := \{\varphi_t(x), t \geq 0\}, \quad \gamma_-(x) := \{\varphi_t(x), t \leq 0\}.$$

(iii) Define the **omega limit set** and the **alpha limit set** as

$$\omega(x) := \left\{y : \varphi_{t_n}(x) \to y \text{ for some } t_n \to \infty\right\},$$
$$\alpha(x) := \left\{y : \varphi_{t_n}(x) \to y \text{ for some } t_n \to -\infty\right\}.$$

(iv) Say a set D is **positively invariant** if for each $x \in D$, $\gamma_t(x) \subset D$ for each $t \geq 0$. Similarly define **negatively invariant** sets.

Example 9.3.

$$x' = \begin{pmatrix} -1 & \\ & -2 \end{pmatrix} x$$

has solution

$$x = \begin{pmatrix} e^{-t} x_{0,1} \\ e^{-2t} x_{0,2} \end{pmatrix}.$$

Thus any ball $\{x_1^2 + x_2^2 < R^2\}$ is positive invariant.

Example 9.4. If $x' = F(x)$ and $F \cdot x > 0$ on ∂B_R, then the solution cannot stay on ∂B_R. To see this, note that $F \cdot x$ is the inner product with the normal vector on the boundary. Alternatively, note that $2F \cdot x$ is the derivate of the Liapunov function $L = |x|^2$. More generally, if

$$\mathbf{n} \cdot F(x)\Big|_{\partial D} < 0,$$

then D is positive invariant.

A similarly conclusion if $F \cdot x < 0$ on ∂B_R.

9.1.1 Some Immediate Properties

Proposition 9.5. φ_t has a semi group property: $\varphi_{t+s} = \varphi_t \circ \varphi_s$.

Proposition 9.6. If x, z lie on the same trajectory, then $\omega(x) = \omega(z)$ and $\alpha(x) = \alpha(z)$.

Proof. Write $x = \varphi_{s_1}(x_0)$ and $z = \varphi_{s_2}(x_0) = \varphi_{s_2 - s_1}(x)$. For any fixed $y \in \omega(x)$, there exists by definition $t_n \to \infty$ such that $\varphi_{t_n}(x) \to y$. Thus $\varphi_{t_n + (s_2 - s_1)}(z) \to y$, giving $\omega(x) \subset \omega(z)$. By symmetry, we have $\omega(x) = \omega(z)$. □

Proposition 9.7. Suppose D is positive invariant and closed. If $x \in D$, then $\omega(x) \subset D$. In particular, $\omega(D) \subset D$.

Proof. Fix $y \in \omega(x)$. Then there exists $t_n \to \infty$ such that $\varphi_{t_n}(x) \to y$. Since D is positive invariant, we have $\varphi_{t_n}(x) \in D$ for each n. Since D is closed, we have $y \in D$. □

Proposition 9.8. $\omega(x)$ is a closed positive invariant set.

Proof. Fix $y \in \omega(x)$. Then there exists $t_n \to \infty$ such that $\varphi_{t_n}(x) \to y$. Note that for each fixed s, we have

$$\varphi_s(y) = \varphi_s\left(\lim_{n\to\infty} \varphi_{t_n}(x)\right) = \lim_{n\to\infty} \varphi_s(\varphi_{t_n}(x)) = \lim_{n\to\infty} \varphi_{t_n+s}(x) \in \omega(x).$$

Thus $\omega(x)$ is positively invariant. To see that $\omega(x)$ is closed, note that it contains all limit points (routine exercise). □

Consider the system

$$x' = F(x), \quad F, x \in \mathbb{R}^{2\times 1},$$

where F is Lipschitz in any bounded domain.

Recall continue dependence implies that for $\varphi_s(x), \varphi_s(y) \in D$, we have

$$|\varphi_t(x) - \varphi_t(y)| \le e^{Lt}|x - y|, \quad \forall t \ge 0,$$

where L is the Lipschitz constant of F on D.

Recall also the implicit function theorem: for $F \in C^1$ with $F(x_0, y_0) = 0$ and $\partial_y F(x_0, y_0) \neq 0$, there exists $y = y(x)$ such that $F(x, y(x)) = 0$ for any x in a small neighborhood of x_0, $B_\varepsilon(x_0)$. Moreover, $y(x)$ is C^1.

Theorem 9.9 (Poincare-Bendixson (Rought Statement)). Suppose $\omega(x)$ is bounded. Then, $\omega(x)$ is either an equilibrium or contains a closed orbit (i.e. a periodic solution). Note that this is not generally true for higher dimensions.

Note that the same result for 1×1 system is easy. The idea of the proof is to reduce the 2D dynamics to 1D, and to prove in a sense that $S \cap \gamma_+(x)$ is monotone, where S is a transversal section.

9.2 Local Sections and Flow Boxes

The context is the system $x' = F(x), x, F \in \mathbb{R}^{2\times 1}$ with F Lipschitz.

Fix a planar system $x' = F(x)$. The **transverse line at x_0, $\ell(x_0)$**, is the straight line through x_0 perpendicular to the vector $F(x_0)$ based at x_0. It is parameterized as follows: Let v_0 be a unit vector perpendicular to $F(x_0)$ and define $h : \mathbb{R} \to \ell(x_0)$ by $h(u) := x_0 + uv_0$.

By continuity, $F(x) \cdot F(x_0) > 0$ (i.e., the two vectors point in the same direction) for each x in a small neighborhood of x_0, say $|x - x_0| < \varepsilon$. Define the **local section** at x_0 by

$$S := \{h(u) \in \ell(x_0), |u| < \varepsilon\} \subset \ell(x_0).$$

Note that each solution starting from S follows in the direction of $F(x_0)$ for small positive time.

Now write $\Psi(s, u) := \varphi_s[h(u)]$ and

$$U_\sigma := \{|u| \text{ small}, |s| < \sigma\}, \quad V_\sigma := \Psi(U_\sigma).$$

The set V_σ is called the **flow box** at x_0. If σ is small enough, then Ψ is one-to-one from U_σ to V_σ (one can compute the Jacobian $|\mathrm{D}\Psi(0, 0)|$ to be nonzero). Thus for each $x \in V_\sigma$, there exists unique $(s, u) \in U_\sigma$ such that $\varphi_s[h(u)] = x$. In particular, we can "pull back" x to the section S by flowing backwards for time s:

$$\varphi_{-s}(x) = \varphi_{-s}(\varphi_s(h(u))) \in S.$$

Proposition 9.10. Suppose $\varphi_{t_0}(z_0) = x_0$ (note that we do not require t_0 to be small). Then there exists an open U containing z_0 and a continuous function $\tau \in C^0(U)$ such that

$$\varphi_{\tau(z)}(z) \in S_{x_0}, \quad \forall z \in U$$

and $\tau(z_0) = t_0$. Note that τ is interpreted as the arrival time to the section S_{x_0}.

Proof. Note that $y \in \ell(x_0)$ if and only if $(y - x_0) \cdot F(x_0) = 0$. Define $G(z,t) = \varphi_\tau(z) \cdot F(x_0) \in \mathbb{R}$. Define $H(z,t) := G(z,t) - x_0 \cdot F(x_0)$. Thus $H = 0$ if and only if $\varphi_t(z) \in \ell(x_0)$. Note that we have $H(t_0, z_0) = [\varphi_{t_0}(z_0) - x_0] \cdot F(x_0) = 0$ by assumption that $\varphi_{t_0}(z_0) = x_0$. Since

$$\begin{aligned}\partial_t H\Big|_{(t_0,z_0)} = \partial_t G\Big|_{(t_0,z_0)} &= \frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(z) \cdot F(x_0)\Big|_{(t_0,z_0)} \\ &= F(\varphi_t(z)) \cdot F(x_0)\Big|_{(t_0,z_0)} = F(x_0) \cdot F(x_0) \neq 0,\end{aligned}$$

by the implicit function theorem, there exists a unique continuous function $\tau(z)$ such that $H(\tau(z), z) = 0$ for each z in a small neighborhood of z_0, and $\tau(z_0) = t_0$. □

9.3 The Poincare Map

The **Poincare map** $P : S_{x_0} \supset D_P \to S_{x_0}$ sends each $x \in S_{x_0}$ to its first return point on $S(x_0)$, if it does return. An application of the implicit function theorem as in the previous proposition guarantees that P is defined and C^1 in a neighborhood of x_0 in S_{x_0}.

The Poincare map is monotone in the following sense:

Proposition 9.11. Suppose that S_{x_0} is a local section and $y_i = \varphi_{t_i}(x) \in S_{x_0}$, where $t_1 < t_2 < t_3 < \ldots$. Then, $y_1, y_2, \ldots$ are monotone along S_{x_0}. That is, y_n is between y_{n-1} and y_{n+1} on S_{x_0} for each n.

Proof. It suffices to show that any three consecutive points y_0, y_1, y_2 are monotone.

Denote as $\Sigma := \{\varphi_t(x), t_0 \le t \le t_1\}$ the arc of the trajectory from y_0 to y_1. Define T to be the segment of S_{x_0} between y_0 and y_1. Define D to be a domain with boundary $\Sigma \cup T$. (Note that $\Sigma \cup T$ is a simply connected domain and cuts the plane into two parts by the Jordan curve theorem.)

Suppose $\varphi_t(x)$ leaves the closure of D at y_1 (the other case of $F(y_1)$ entering D is similar, with D being positive invariant). In this case D^c must be positive invariant, since any trajectory starting from D^c cannot enter D by crossing Σ as trajectories cannot intersect; nor can it cross T due to the direction of the flow there.

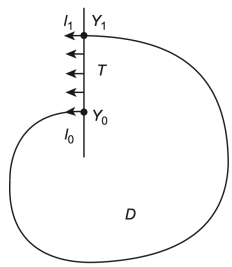

Figure 10.7 Solutions exit the region D through T.

Write $S = T \cup I_0 \cup I_1$ (see figure above). We claim that $I_1 \subset D^c$ and $I_0 \in D$. To see this, suppose for contradiction that $I_1 \subset D$. Pick a point $p \in I_1 \subset D$ close to y_1. The segment from the point $\varphi_\varepsilon(y_1) \in D^c$ (for some small $\varepsilon > 0$) to the point $p \in D$ must cross ∂D at some point. But this point cannot be on T since both endpoints of the segment is on the same side of S. Thus we can write the point of intersection as $\varphi_{-\beta}(y_1)$ for some $\beta > 0$. That is,

$$\varphi_{-\beta}(y_1) = \lambda\varphi_\varepsilon(y_1) + (1-\lambda)p$$

and so

$$\varphi_{-\beta}(y_1) - y_1 = \lambda[\varphi_\varepsilon(y_1) - y_1] + [(1-\lambda)p - y_1].$$

Now take the dot product of each term with $F(x_0)$. The third term vanishes, and the remaining two terms have different signs (being on different sides of S). This is clearly impossible.

Now, since y_2 must stay in D^c and cross S not on T (by directions of the flow on T), it must be on I_1. This gives the desired monotonicity. □

Proposition 9.12. Suppose $y \in \omega(x)$. The solution through y crosses any S at most once.

Proof. Suppose for contradiction that the solution through y crosses S at least twice, say at $y_1 = \varphi_{t_1}(y)$ and $y_2 = \varphi_{t_2}(y)$, where $t_1 < t_2$. We can draw small flow boxes V_1, V_2 around y_1 and y_2 that does not intersect. Since $\omega(x)$ is positive invariant, $y_1, y_2 \subset \omega(x)$. Thus $V_1 \ni \varphi_{t_n}(x) \to y_1$ and similarly $V_2 \ni \varphi_{s_n}(x) \to y_2$ for some sequences $t_n, s_n \to \infty$. We can shift time slightly to land on the local section:

$$\varphi_{t_n+o(\sigma)} \in S, \quad \varphi_{s_n+o(\sigma)} \in S.$$

Thus there exists large k and n such that

$$t_{1+o(\sigma)} < s_{k+o(\sigma)} < t_{n+o(\sigma)},$$

which contradicts the monotonicity of the Poincare map. □

9.4 Poincare-Bendixson

Theorem 9.13 (Poincare-Bendixson). Suppose $\omega(x)$ is nonempty, closed, bounded, and does not contain an equilibrium point. Then, $\omega(x)$ is a closed orbit.

Proof. We know $\omega(x) \subset K$ is nonempty (by K being compact), closed (standard property), bounded (inherited from K), and positive (negative) invariant. Thus we can pick a $y \in \omega(x)$. Note that $\varphi_t(y) \in \omega(x)$ for each t, which implies that $\omega(y) \subset \omega(x)$. Again by compactness, $\omega(y) \neq \emptyset$. Thus fix $z \in \omega(y) \subset \omega(x) \subset K$. We have $F(z) \neq 0$ by the assumption that $\omega(x)$ contains no equilibrium point.

Pick a local section S_z and a flow box V_σ of z. We have $\varphi_{t_n}(y) \to z$ for some $t_n \to \infty$. For large n we have $\varphi_{t_n}(y) \in V_\sigma$ and so can shift time slightly to land on the section:

$$\varphi_{t_n+o(\sigma)}(y) \in S_z.$$

By the previous proposition, these points are all the same. Hence y is periodic. Now let γ be the closed orbit through y. Since $y \in \omega(x)$, we have that $\gamma \subset \omega(x)$ by positive invariance.

We next show $\omega(x) \subset \gamma$. Note that it is sufficient to establish that

$$\lim_{t\to\infty} d\left(\gamma, \varphi_t(x)\right) = 0,$$

where d is the obvious generalization of distance between a point and a set. There is some sequence $t_1 < t_2 < \ldots$ such that

$$\varphi_{t_n}(x) \to y, \quad \varphi_{t_n}(x) \in S_y,$$

and for any $t \in (t_n, t_{n+1})$,

$$\varphi_t(x) \notin S_y.$$

The inter-arrival times $t_{n+1} - t_n$ are bounded above $\tau + \varepsilon$, where τ is the period of the closed orbit and ε is the size of the flow box (the size of its preimage under Ψ). By continuous dependence on initial conditions, for each $t \in (t_n, t_{n+1})$, we have

$$d(\gamma_t, \varphi_{t+t_n}(x)) = d(\varphi_t(y), \varphi_{t+t_n}(x)) \leq C|\varphi_{t_n}(x) - y|.$$

The last term goes to zero as $n \to \infty$. □

Corollary 9.14. Suppose K is closed, bounded, positive (negative) invariant. Then K contains an equilibrium or a closed orbit.

Proof. Pick $x \in K$. It is easy to check that $\emptyset \neq \omega(x) \subset K$. If $\omega(x)$ contains no equilibrium point, then by Poincare-Bendixson theorem, $\omega(x)$ is a closed orbit. □